Explain the Working principle of Arc welding with neat sketch.
Describe the construction & working of Arc welding process with neat sketch

Arc welding: It is a type of fusion welding. It operates under the principle that when two conductors of an electric circuit are touched together momentarily and then instantaneously separated slightly, assuming that there is sufficient voltage in the circuit to maintain the flow of current, an electric arc is formed. Concentrated heat is produced throughout the length of the arc at a temperature of about 5000 to 6000°C. In arc welding, usually the parts to be welded are wired as one pole of the circuit, and the electrode held by the operator forms the other pole. When the arc is produced the intense heat quickly melts the work piece metal which is directly under the arc, forming a small molten metal of the electrode is carried over by the arc to the molten metal pool of the work piece. The molten metal in the pool is agitated by the action of the arc, thoroughly mixing the base and the filler metal. A solid joint will be formed when the molten metal cools and solidifies. The flux coating over the electrode produces an inert gaseous shield surrounding the arc and protects the molten metal from oxidizing by coming in contact with atmosphere.

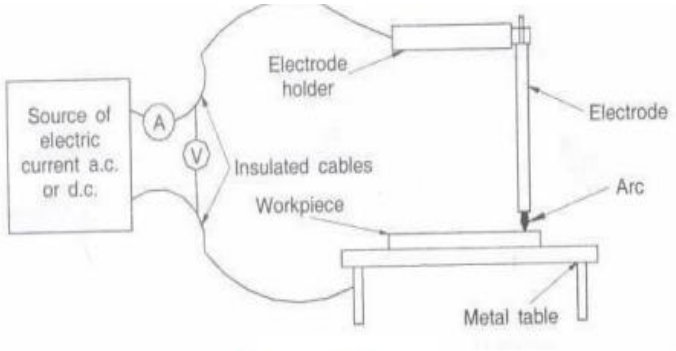
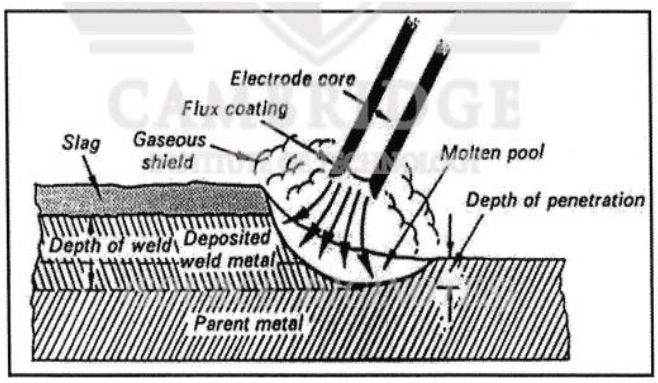


Figure: Arc Welding Setup



Explain the different types of flames used in Gas welding.

(i) Neutral flame:

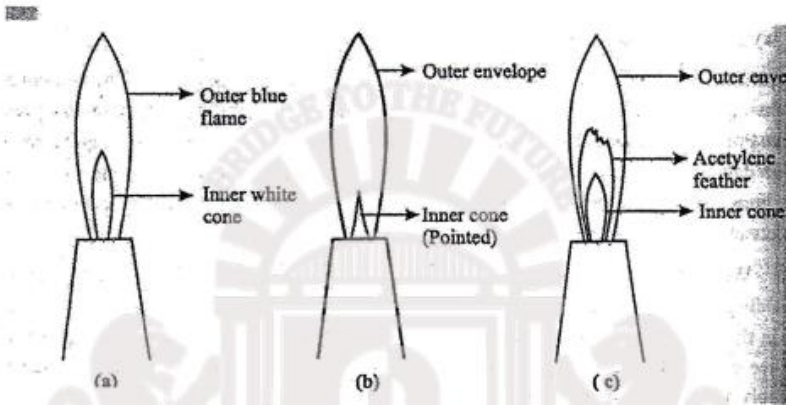
- A neutral flame is obtained by supplying equal volume of oxygen and acetylene
- It consists of a small whitish inner cone surrounded by sharply defined blue flame
- Most of the gas welding is done using the neutral flame

(ii) Oxidizing flame

- This is obtained when there is excess of oxygen, gas ratio
- It appears to be similar to that of neutral flame but the inner white cone flame is shorter than that of neutral flame
- This flame is generally used in metal cutting rather than welding since weld metal gets oxidized

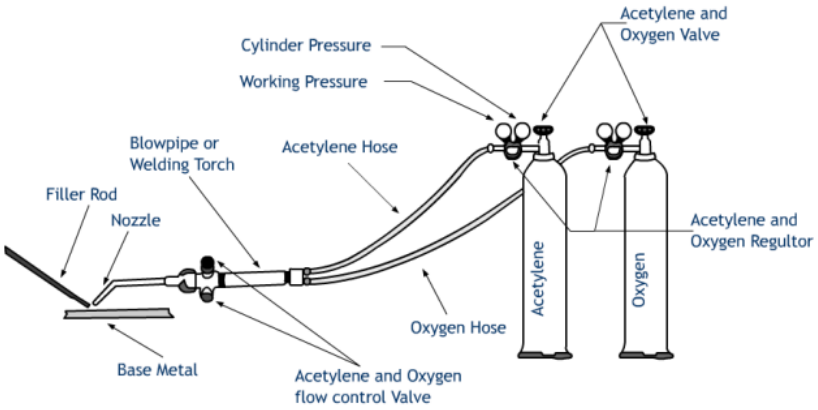
(iii)Reducing flame:

- This is obtained by supplying excess of acetylene in the gas ratio
- It has 3 cones, an inner white cone ,surrounded by an intermediate whitish cone known as " intermediate flame feather" and a bluish envelope flame
- This flame is used for welding alloy steels, cast iron, aluminium



With the help of a neat sketch explain the working of Gas welding

Oxy-acetylene welding: It is a fusion welding, in which a strong gas flame is used to raise the temperature of the work piece to melt them. As in the arc welding, a filler metal is used to fill the joint. The gases that can be used for heating are (i) Oxygen and acetylene (ii) Oxygen and hydrogen. Oxy-acetylene gas mixture is most used in gas welding. When right proportions of oxygen and acetylene are mixed in the welding torch flame produced is called as the oxy-acetylene flame. The temperature attained in 3200° C, hence has an ability to melt all commercial metals.



Enumerate the Difference between Soldering, Brazing and Welding.

Comparison between Welding, Soldering and Brazing:

Welding	Soldering	Brazing
It is a high temperature process where in the base metals are heated above their melting temperature	Low temperature process, base metals are not melted	The base metals are not melted but broadly heated to a suitable temperature
Filler metals is same as that of base metals	Filler metals is not the same as that of base metals	Filler metals is not the same as that of the base metals
Joint is formed by the solidification of the molten filler metal with the molten base metal	Joints is formed by means of diffusion of the filler metal into the base metal	Joint is formed by means of diffusion of the filler metal into the base metal associated with surface alloying
Strength of weld is much stronger	Strength is low	Lies between welded and soldered joint

Due to high temperature the metal adjacent to the weld portion called heat affected zone is affected to a larger extent	There is no Heat affected zone as the process is carried out at low temperature	Although base metals are heated, the heat affected zone is not too much when compared to welding
Welded joint requires finishing operations	Finishing operation is not required	Little finishing is required
Temperature >3000°C	Temperature <450°C	Temperature >450°C and less than melting point of base metal

What is Soldering? Explain the process of soldering

Soldering

• Soldering is a method of joining similar or dissimilar metals by the application of heat and using a filler metal or alloy called solder, whose liquidus temperature is below 450° C.

• The molten filler metal is made to flow between the two closely placed adjacent surfaces by the capillary action.

• Though soldering obtains a good joint between the two plates, the strength of the joint is limited by the strength of the filler metal used.

Sequence of operations: The following operations are required to be performed sequentially for making soldered joints.

• Shaping and fitting of metal parts together: The two parts to be joined are shaped to fit closely so that the space between them is extremely small and filled completely with solder by capillary action. If a large gap is present, capillary action will not take place, and the joint will not be strong.

• Cleaning of surfaces: In order to obtain a sound joint, the surfaces to be soldered are cleaned to remove dirt grease or any other foreign material.

• Application of flux: The flux is applied when the parts are ready for joining.

• Application of heat and solder: The parts are held in a vice or with special work holding devices so that parts do not move while soldering.

Advantages:

• Simple and economical process.

• Relatively low temperature process, there is no metallurgical damage to base metal.

• The soft soldered joints can easily be dismantled by simple heating.

Disadvantage:

• The strength of joint relatively low.

• Flux must be thoroughly cleaned off after soldering, as it is often corrosive.

What is Brazing? Explain the process of soldering

Brazing is a process of making joints where in coalescence is produced by heating to suitable temperatures above 500° C and by using a non-ferrous filler metal having a melting point (up to 900° C) below that of the base metal, the filler metal being distributed between the closely fitted surfaces of the joint by capillary action. Brazing gives a much stronger joint than soldering

The principal difference is the use of a harder filler material commercially known as spelter. Filler metals used in this process may be divided into copper base alloys and silver base alloys. The spelter is usually an alloy of copper, zinc and tin. Both similar and dissimilar metals can be joined. • The flux along with spelter (filler metal) is applied to remove oxides from the surfaces. Borax is the most widely used flux. It will dissolve the oxides of most of the common metals. Other fluxes used are mixtures of borax, boric acid, fluorides and chlorides.

Advantages:

- Dissimilar metals, such as stainless steel to cast iron can be joined by brazing. Almost all metals can be joined by brazing except aluminium and magnesium which cannot easilybe joined by brazing.
- Because of the lower temperatures used there is less problems due to heat.
- The joint can be quickly finished without much skill.
- Because of the simplicity of the process it is often an economical joining method with reasonable joint strength.
- The brazed joints are reasonably stronger, depending on the strength of the filler metal used.

Disadvantages:

- Flux material required to prevent the corrosion.
- Large section cannot be joined.
- Fluxes and filler materials may toxic.
- Filler materials are expensive

Describe the following Materials i) Ceramics ii) Shape memory Alloys

Describe the three types of carbon steels with applications

Low Carbon Steel: It is also called Mild steel and the most common type of carbon steel. It contain 0.03-0.25%off carbon and rest is iron It is soft, ductile and it has good machinability, good weldability, good toughness, good impact resistance.

Applications: It finds applications in nails, rivets, bolts, nuts, beams, boiler plates, shafts, gears and axles.

Medium Carbon Steel: It has 0.25-0.59% of carbon and rest is iron

Properties: like good toughness, high hardness, high tensile strength, good bending strength, wear resistance and good machinability.

Applications: It is used in transmission shafts, axles, gears, connecting rods

Alloy Steel : An alloy can be defined as a homogeneous mixture of two or more metals. Alloy steel include iron and carbon.

Stainless Steel • It is an iron-carbon alloy with chromium content of 11.5%-25%depending upon the properties • Chromium helps to resist corrosion in steels. • They get the name because they do not corrode or stain easily.

Applications: Manufacturing of kitchen utensils, sinks, food processing equipment, chemical plant and equipment, knife blades, surgical instruments etc.,

Chromium Steel • In this type of steel chromium is the alloying element • It improves toughness, resistance to heat, corrosion

Applications: application in balls, rollers, cutting tools

Write a short note on Shape Memory Alloys

Shape-memory alloy: • Shape-memory alloy (SMA) is an alloy that can be deformed when cold but returns to its predeformed ("remembered") shape when heated. • It may also be called memory metal, memory alloy, smart metal, smart alloy, or muscle wire. • Parts made of shape-memory alloys can be lightweight, solid-state alternatives to conventional actuators such as hydraulic, pneumatic, and motor-based systems. • They can also be used to make hermetic joints in metal tubing.

Types of Shape-memory alloy • The two most prevalent shape-memory alloys are copper-aluminium-nickel and nickeltitanium (NiTi), but SMAs can also be created by alloying zinc, copper, gold and iron. • Although iron-based and copper-based SMAs, such as Fe-Mn-Si, Cu-Zn-Al and Cu-Al-Ni, are commercially available and cheaper than NiTi, NiTi-based SMAs are preferable for most applications due to their stability and practicability as well as their superior thermo-mechanic performance.

Applications of Shape-memory alloy:

• Recent research of SMA has been applied in the field of Aerospace, Automotive, Biomedical, and Robotics. • This memory effect is due to the presence of austenite and martensite crystalline structures. These alloys are bio-compatible, lightweight, and also possess a high force-to-weight ratio.

Defrine the following terms:

i. Ceramic ii. Graphite iii. Polymers

Ceramics

• Ceramics are inorganic solid materials containing compounds of metallic and non-metallicelements most frequently oxides, nitrides and carbides • Ceramics are good for high temperature applications mainly because of their unique mechanical, chemical, structural and functional properties • Example: Clay, Bricks, Tiles, Glass, Cement

Graphite:
• Graphite is a crystalline form of the element carbon. It consists of stacked layers of graphene. • Graphite occurs naturally and is the most stable form of carbon under standard conditions. • Synthetic and natural graphite are consumed on large scale for uses in pencils, lubricants, and electrodes. • Under high pressures and temperatures, it converts to diamond. It is a weak conductor of heat and electricity.

Polymers: • These are the organic materials with carbon as a element in their formation • Ploymers are composed of large number of small molecules called manomers • Examples: Polyethylene, polyvinyl chloride, Nylon, Polypropylene, Polythene

Properties: • They have low density • They exhibit good corrosion resistance • They process good mouldability • They have low coefficient of friction • They give excellent surface finish • Low mechanical properties

Define rhe engineering materials? Explain the classifications of Ferrous and Non ferrous materials

Explain the different types of Ferrous materials with applications.
Explain the different types of Nonferrous materials with applications

OR

Recite the classification of metals

Engineering materials are defined as solid substances which are manufactured and used for various engineering applications.

Pig Iron: It is the first stage of iron directly extracted from the ore through blast furnace. It Contain high percentage of carbon and other impurities.
Properties: High hardness and Vey brittle
Applications: It is used in making wrought iron, cast iron and steel

Wrought iron: Wrought iron is an iron alloy with a very low carbon content
Properties: Low carbon content, Soft, Highly Ductile in nature, High toughness, Good machinability, Corrosion resistance, High elasticity.
Applications: Due to corrosion resistance it is used in making steam and water pipes. Decorative applications like fences, gates and railings, balconies, bars, engine bolts, rivets, railway couplings, chain links and crane hooks.

Cast Iron: Is a ferrous material that is produced when pig iron is remelted in a cupola furnace .

Properties: Very strong but brittle, Low melting point, Good fluidity, Castability, Goodmachinability, wear resistance

Applications: Manufacture of machine frames, columns, beds and plates, housing, flywheels, manhole covers, engine block, cylinder head, gearbox case, machine parts etc,

Grey Cast: Iron It is produced when the molten pig iron in the mould is allowed to cool and solidify slowly. Carbon here is mainly in the form of graphite

Properties: Good castabality, High fluidity, easy Machinability, high strength and ductility under compressive loads. It is named as Grey cast iron because of its freshly fractured surface has a grey appearance

Applications: Base structures of machinery, Automotive engine blocks, gears, flywheels, brake discs and drums and machine bases.

STEEL

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Non-Ferrous Metals

	These are metals without iron content. These materials are preferred due to • Good strength to weight ratio • Good resistance to corrosion • Light weight • High electrical and thermal conductivity • Ease fabrication **Aluminium** • It is extracted from Bauxite • The mechanical properties can be improved by mixing aluminium with silica, copper,zinc **Properties:** • Light weight, high ductility, good electrical and thermal conductivity, good resistance tocorrosion **Application:** • Used in making aircraft parts • Used in automobile for making cylinder block, piston and other engine componenets • Used in domestic application for making cooking utensils • Electrical industry- cables, induction motor, conductors rotors Alloys of Aluminium- Duralumin, Magnalumin **Copper** • Reddish brown colour, extracted from PYRITE • It is soft, malleable, ductile, light weight, high thermal conductivity, good corrosionresistance, good electrical and thermal conductivity Appli**cations**: • Used to make door knobs • Used in electrical wires, cable, conductors • Used to make alloy of brass and bronze • In the form of tubes used in refrigerator, air conditioner and radiator due to high thermalconductivity **Alloys of Copper:** Brass, Bronze, Gun Metal **Lead** Alloys of Lead • Solder: Lead and tin used for joining less fusible metals Lead foil: is made sandwich of lead between two sheets of tin,. Lead foils are used inradiation shielding • Lead-antimony alloy: are used in storage batteries grid Zinc • Is heavy bluish white metal which can be extracted from Zinc Sulphide or Zinc blend • It is fair conductor f electricity • It has low melting point and boiling point • It is resistance to corrosion and good castability Application • Galvanizing: To protect iron and steel form corrosion, they are immersed in a bath ofliquid zinc or electroplated with zinc, this is called galvanizing • Galvanised steels is used in produce automobile parts Alloy of Zinc • Cadmium zinc alloy • Magnesium zinc alloy • Copper zinc alloy • Lead zinc alloy • Iron zinc alloy